# KAPREKAR CONTEST – FINAL – SUB JUNIOR

AMTI – OCTOBER, 2004.

Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Attach the filled-in FACE SLIP and your rough working sheets along with the answer script.
5. Maximum time allowed is THREE hours.

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1. The greatest common divisor of $a$ and 72 is $(a,72)=24$ and the least common multiple of $b$ and 24 is $[b,24]=72$. Find the g.c.d $(a,b)$ and the l.c.m $[a,b]$ given that $a$ is the smallest three digit number having this property; and $b$ is the biggest integer having this property.

2. Given that $a^2 - b^2 = 105$ and $a$ and $b$ are two relatively prime positive integers (two positive integers $m$ and $n$ are relatively prime if their g.c.d $(m,n)=1$), find all such $a$ and $b$. After having found all such $a$ and $b$, if one draws a triangle $ABC$ with sides having lengths $a^2 - b^2$, $a^2 + b^2$ and $2ab$ find the area of all such triangles.

3. $A$ is a set of 2004 positive integers. Show that there is a pair of elements in $A$ whose difference is divisible by 2003.

4. Let $ABC$ be an acute angled triangle with $AD$, $BE$, $CF$ as the altitudes (i.e., $D$ is the foot of the perpendicular from $A$ on $BC$ and so on . . . ). If the altitudes meet at the point $O$, find the measure of the angles $\angle BOC$, $\angle COA$, $\angle AOB$ in terms of the angles $\angle A$, $\angle B$, $\angle C$ of the triangle $ABC$.

5. Is the statement "If $p$ and $p^2 + 2$ are primes then $p^3 + 2$ is also a prime" true or false? Give reasons for your answer.

6. Let $P$ denote the product of first $n$ prime numbers (with $n>2$). For what values of $n$ we have
   a) $P - 1$ is a perfect square
   b) $P + 1$ is a perfect square

7. Three counters $A$,$B$ and $C$ are coloured with three different colours red, blue and white. Of the following statements only one is true.
   a) A is red.
   b) B is not red.
   c) C is not blue.
   What is the colour of each counter?

8. The sum and least common multiple of two positive integers $x,y$ are given as $x+y = 40$ and l.c.m $[x,y] = 48$. Find the numbers x and y.

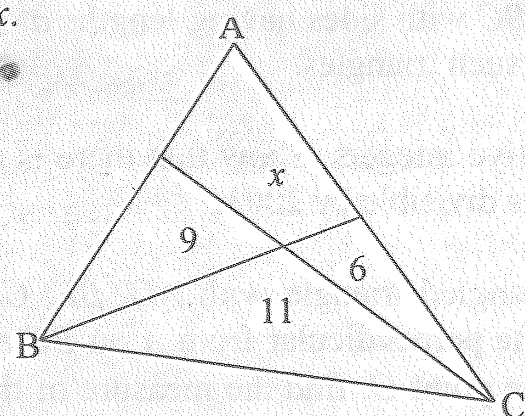
9. What is the greatest positive integer $n$, which makes $n^3+100$ divisible by $n+10$?

10. Find the sum of all three digit numbers that can be written using the digits 1,2,3,4 (repetitions allowed).

11. Consider the collection $C$ of all isosceles triangles of area 48 sq.units, whose bases and heights are integers. How many triangles are there in $C$? How many triangles in $C$ have their equal sides also of integral lengths?

12. In triangle $ABC$, we are given that $\angle A=90^\circ$. Median $AM$, angle bisector $AK$ and the altitude $AH$ are drawn. Prove that $\angle MAK = \angle KAH$.

13. Triangle $ABC$ is divided into four regions with areas as shown in the diagram. Find $x$.




14. $ABCD$ is a cyclic quadrilateral (which means that a circle passes through the vertices $A,B,C,D$).In other words the vertices $A,B,C,D$, in that order, lie on a circle. If the diagonals $AC$ and $BD$ cut at right angles at $E$, prove that $AE^2 + BE^2 + CE^2 + DE^2 = 4R^2$ where $R$ is the radius of the circle $ABCD$.

15. $A= \{a,b,c,d,e\}$ is a set of five integers. We take two out of the numbers in $A$ and add. The following ten sums are obtained

$$0,6,11,12,17,20,23,26,32,37.$$

Find the five integers in the set A.